Week 11 Workshop

Question 1
Consider the GARCH(1,1) model

$$\begin{aligned} r_t &= \mu + \varepsilon_t \\ \varepsilon_t &= \sigma_t u_t \qquad u_t \sim iid\ N(0,1) \\ \sigma_t^2 &= \alpha_0 + \alpha_1 \varepsilon_{t-1}^2 + \beta_1 \sigma_{t-1}^2 \end{aligned}$$

Assume $\mu = 0$. Compare the following three models.

$$\begin{aligned} \text{Model A} &: \alpha_0 = 0.001, \alpha_1 = 0.1, \beta_1 = 0.85 \\ \text{Model B} &: \alpha_0 = 0.010, \alpha_1 = 0.1, \beta_1 = 0.85 \\ \text{Model C} &: \alpha_0 = 0.050, \alpha_1 = 0.1, \beta_1 = 0.85 \end{aligned}$$

a. Calculate $\alpha_1 + \beta_1$ for each model.

b. Do the three models have the same level of volatility persistence?

c. Derive the unconditional variance.

d. Calculate the unconditional variance for each model.

e. Which model is expected to generate returns with the largest sample variance?

f. Simulate $T = 1000$ observations from each model. Use the unconditional variance as the initial value for σ_1^2. Calculate the sample variance of the simulated returns from each model.

g. Compare the sample variance with the theoretical unconditional variance. Explain why the simulated sample variance may not be exactly equal to the theoretical value.

h. Does increasing α_0 necessarily increase volatility clustering?

i. Does a larger sample variance necessarily mean stronger volatility clustering?

Compare the following two models

$$\begin{aligned} \text{Model D} &: \alpha_0 = 0.10, \alpha_1 = 0.1, \beta_1 = 0.40 \\ \text{Model E} &: \alpha_0 = 0.01, \alpha_1 = 0.1, \beta_1 = 0.85 \end{aligned}$$

j. Show that both models have the same unconditional variance.

k. Which model has stronger volatility persistence?

l. Explain why this example illustrates the usefulness of GARCH models.

Question 2

Forecasting the Volatility of Australian Stock Returns. Consider the daily data on the Australian stock price index (PRICE) from 24 February 2003 to 30 September 2019. Let r_t denote the log return.

(a) Write down the fitted model corresponding to the results in the following Figure. The residual ($\hat{\varepsilon}_t$) and the fitted conditional variance ($\hat{\sigma}_t^2$) on 30 September 2019 are $\hat{\varepsilon}_t = -0.003992$ and $\hat{\sigma}_t^2 = 3.13 \times 10^{-5}$. Forecast the variance for October 1, 2019.

(b) Use the parameter estimates to compute $var(\varepsilon_t)$.

```
> library(rugarch)
> spec_garch <- ugarchspec(
+   variance.model = list(
+     model = "sGARCH",
+     garchOrder = c(1, 1)
+   ),
+   mean.model = list(
+     armaOrder = c(0, 0),
+     include.mean = TRUE
+   ),
+   distribution.model = "norm"
+ )
>
> fit_garch <- ugarchfit(
+   spec = spec_garch,
+   data = log_ret
+ )
>
> fit_garch@fit$matcoef
           Estimate   Std. Error    t value     Pr(>|t|)
mu     5.427819e-04 1.012755e-04  5.3594602 8.347099e-08
omega  7.757249e-07 1.177561e-06  0.6587556 5.100527e-01
alpha1 9.179722e-02 2.411360e-02  3.8068658 1.407391e-04
beta1  9.009662e-01 2.368101e-02 38.0459330 0.000000e+00
```